§2.3 Polynomial Functions of Higher Degree with Modeling

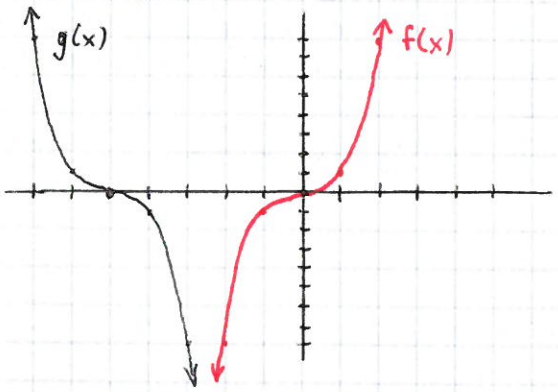
208-212 / Q1-Q10, 1-7 odd, 9-12, 13-27 odd, 29-32, 33-55 odd, 69-74

Recall: A polynomial fcn of degree 0 is a constant fcn and graphs as a horizontal line
A polynomial fcn of degree 1 is a linear fcn; its graph is a slant line
A polynomial fcn of degree 2 is a quadratic fcn; its graph is a parabola

A polynomial fcn of degree n can be written $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$, $a_n \neq 0$

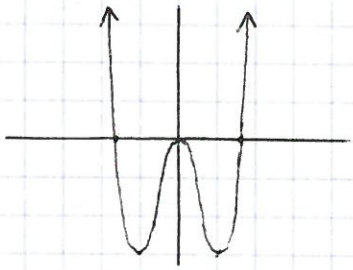
- Each monomial is a <u>term</u>: e.g. $a_n x^n$, $a_2 x^2$, a_0
- <u>Standard form</u>: Terms written in descending degree
- <u>Coefficients</u>: The constants a_n, a_{n-1}, ..., a_0
- <u>Leading Coefficient</u>: The constant a_n
- <u>Leading Term</u>: $a_n x^n$

Try p 209 # 2 $g(x) = -(x+5)^3$
Let $f(x) = x^3$. $g(x)$ is a reflection of $f(x)$ over the x-axis and a horizontal translation bafo -5.



y-intercept:
$g(0) = -(0+5)^3$
$= -125$
$(0, -125)$

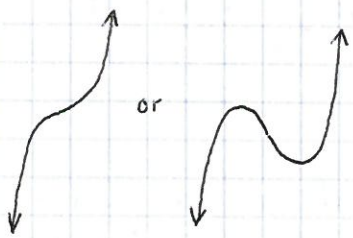
#8 $g(x) = 2x^4 - 5x^2$



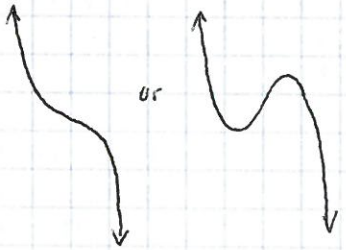
$= x^2(2x^2 - 5)$
zeros at $x = 0$ (double zero) and $\pm\sqrt{5/2} = \pm 1.5811388\ldots$
local minima of -3.125 at $x = \pm 1.11803\ldots$
local maximum of 0 at $x = 0$

<u>Typical Cubic Functions</u>

$a_3 > 0$



$a_3 < 0$



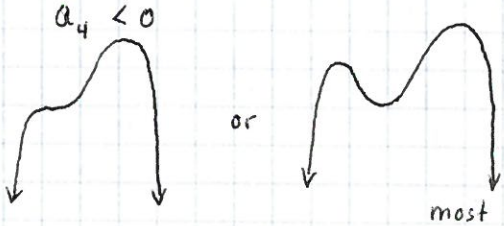
* Cubic functions can have at least one real zero and at most three real zeros (and at most 2 local extrema)

<u>Typical Quartic Functions</u>

$a_4 > 0$



$a_4 < 0$



* Quartic functions could have no real zeros and at most four real zeros (at most 3 local extrema)

Theorem: A polynomial function of degree n has at most $n-1$ local extrema and at most n zeros

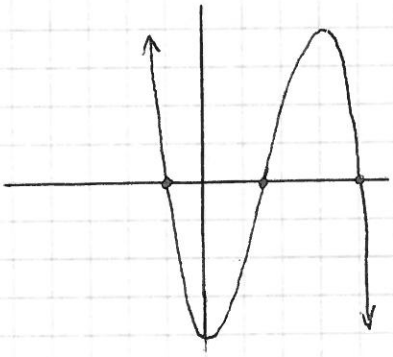
* End behavior of a polynomial is closely related to the end behavior of its leading term. The leading term dominates the behavior of the polynomial as $|x| \to \infty$

#14. $f(x) = x^3 + 2x^2 - x + 5$ and $g(x) = x^3$

Started Zoom 4
Zoomed Out twice
$[-105.6, 105.6]$ by $[-65.6, 65.6]$

See p 204 for Leading Term Test for Polynomial End Behavior

#18 $f(x) = (2x-3)(4-x)(x+1)$



Zeros @ $x = 3/2, 4$, and -1
Local minimum of $-12.0281...$
Local maximum of $12.02813...$

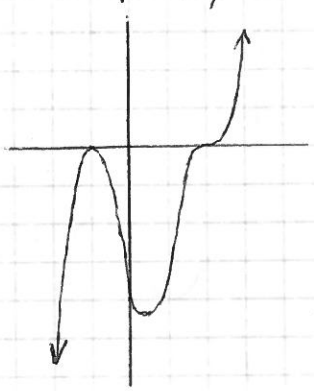
$\lim_{x \to \infty} f(x) = -\infty$ $\lim_{x \to -\infty} f(x) = \infty$

#28 $f(x) = x^3 - x^4 + 3x^2 - 2x + 7$
$f(x) = -x^4 + x^3 + 3x^2 - 2x + 7$
$\lim_{x \to \infty} f(x) = -\infty$ $\lim_{x \to -\infty} f(x) = -\infty$

* End behavior-wise, $f(x)$ will behave the same as $g(x) = -x^4$

#34 $f(x) = 3x^2 + 4x - 4$
$3x^2 + 4x - 4 = 0$
$(3x-2)(x+2) = 0$
$3x - 2 = 0$ $x + 2 = 0$
$x = 2/3$ $x = -2$
$f(x)$ has zeros at -2 and $2/3$

When a factor is repeated, as in $f(x) = (x-2)^3(x+1)^2$, $f(x)$ has a repeated zero. The function shown has two repeated zeros. Because the factor $(x-2)$ is repeated three times, 2 is a zero of <u>multiplicity</u> 3. Similarly, -1 is a zero of multiplicity 2.



even
At a multiplicity of 2, the graph of f is tangent to the x-axis at the zero, $(-1, 0)$. i.e. the graph of f just kisses the x-axis without crossing it.

odd
At a multiplicity of 3, the graph of f crosses through the x-axis at the zero, $(2, 0)$. However, it briefly flattens out horizontally along the x-axis as it crosses through.

#42. $f(x) = 7(x-3)^2(x+5)^4$

degree is 6

zeros at 3 and -5

the zero of 3 (multiplicity of 2) - graph is tangent to the x-axis

the zero of -5 (multiplicity of 4) - graph is tangent to the x-axis

$a_6 > 0$



Intermediate Value Theorem

If a and b are real numbers with $a < b$ and $f(x)$ is continuous on $[a,b]$, then f takes on every value between $f(a)$ and $f(b)$.

i.e. if y_0 is between $f(a)$ and $f(b)$, then $y_0 = f(c)$ for some number $x = c$ in $[a,b]$

∴ if $f(a)$ and $f(b)$ have opposite signs, then $f(c) = 0$ for some number $x = c$ in $[a,b]$ $\Rightarrow$ a sign change guarantees a real zero

#46 $f(x) = -x^4 - 3x^3 + 7x^2 + 2x + 8$



Zeros at $x \approx -4.5$ and $x = 2$

(actual $x = -4.53284...$ and $x = 2$)

#52 $f(x) = x^3 - 4x^2 - 44x + 96$

	1	-4	-44	96
1	1	-3	-47	49
-1	1	-5	-39	135
2	1	-2	-48	0

$f(x) = (x-2)(x^2 - 2x - 48)$

$= (x-2)(x-8)(x+6)$

Let $f(x) = 0$

zeros at $x = -6$, 2, and 8

#54 zeros at $x = -2$, 3, and -5

$f(x) = (x+2)(x-3)(x+5)$

$= (x^2 - x - 6)(x+5)$

$= x^3 + 5x^2 - x^2 - 5x - 6x - 30$

$= x^3 + 4x^2 - 11x - 30$

